Vienna mapping functions in VLBI analyses

Johannes Boehm and Harald Schuh

University of Technology, Vienna, Austria

Received 4 November 2003; revised 2 December 2003; accepted 4 December 2003; published 2 January 2004.

[1] In the past few years, numerical weather models (NWM) have been investigated to improve mapping functions which are used for tropospheric delay modeling in VLBI and GPS data analyses. The Vienna Mapping Functions VMF are based on direct raytracing through NWM, and so they are able to exploit the full information provided in the NWM. On the other hand, the Isobaric Mapping Functions IMF are using intermediate parameters calculated from the NWM. In this study, pressure level data from ECMWF (European Centre for Medium-Range Weather Forecasts) are applied to determine the coefficients of the VMF and the IMF. Used for the analyses of IVS-R1 and IVS-R4 VLBI sessions, both mapping functions improve the repeatability of baseline lengths (by ∼10% for IVS-R1 and ∼5% for IVS-R4) compared to the Niell Mapping Functions NMF. *INDEX TERMS:* 1229 Geodesy and Gravity: Reference systems; 1243 Geodesy and Gravity: Space geodetic surveys; 1247 Geodesy and Gravity: Terrestrial reference systems; 1299 Geodesy and Gravity: General or miscellaneous; 6964 Radio Science: Radio wave propagation. **Citation:** Boehm, J., and H. Schuh (2004), Vienna mapping functions in VLBI analyses, *Geophys. Res. Lett.*, *31*, L01603, doi:10.1029/2003GL018984.

1. Introduction

[2] Raytracing through radiosonde data has often been applied to develop and validate mapping functions which are needed for tropospheric modeling in VLBI and GPS data analyses. For example, the Niell Mapping Functions NMF [*Niell*, 1996], which require station height, station latitude and day of the year as input parameters, were developed using radiosonde data over a wide range of latitudes.

[3] In recent years, much effort has been put into the development of mapping functions which are based on data from numerical weather models. *Niell* [2001] proposed the Isobaric Mapping Functions (IMF) which apply as input parameters the height of the 200 mbar pressure level (‘z200’) and the ratio of the wet path delay along a straight line at 3.3° elevation and its zenith delay (‘smfw3’). The equations relating these two parameters to the coefficients of the continued fraction form (see Equation 1) are based on raytracing through radiosonde data. IMFh provides the information to allow correction for hydrostatic north-south and east-west gradients before estimating the remaining wet gradients.

[4] When working on the implementation of these mapping functions with pressure level data from the ECMWF, it became evident that the NWM could be exploited more rigorously by discarding intermediate parameters like z200 and smfw3. The main idea for the Vienna Mapping Functions VMF is to simply use the raytracing through the NWM directly instead of taking intermediate steps.

2. Determination of the Vienna Mapping Functions (VMF)

[5] The continued fraction form [*Marini*, 1972] for the hydrostatic and wet mapping function for an elevation angle e is shown in equation (1) [*Herring*, 1992]. This form is also used in the NMF [*Niell*, 1996] and in the IMF [*Niell*, 2001].

$$\mathrm{mf(e)} = \frac{1 + \frac{a}{1 + \frac{b}{1+c}}}{\sin e + \frac{a}{\sin e + \frac{b}{\sin e + c}}} \qquad (1)$$

Three coefficients a, b and c are sufficient to map zenith delays down to elevations of 3°. In the case of VMF, these coefficients are determined from raytracing through NWM. A description of the raytracing can be found in *Boehm and Schuh* [2003]. Input parameters for the raytracing program are an initial elevation angle e_0, and values for height, temperature and water vapor pressure at distinct pressure levels in the neutral atmosphere (e.g., 15 levels from 1000 to 10 hPa total pressure). The raytracing then yields the outgoing (vacuum) elevation angle e (see Figure 1), and the values for the hydrostatic and the wet mapping function. The hydrostatic mapping function includes the geometric bending effect.

[6] Two ways of determining the coefficients in equation (1) from raytracing through ECMWF pressure levels will be presented here. The first one is rigorous, whereas the second one is faster and still of equivalent accuracy.

2.1. Rigorous Determination of the Coefficients: VMF(rig)

[7] For each site (e.g., VLBI station) and each epoch when ECMWF pressure level data are available, i.e., every

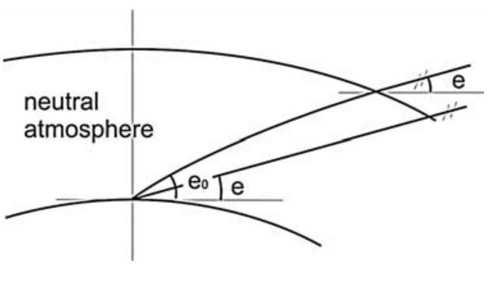

Figure 1. Bending of the ray in the neutral atmosphere. The outgoing (vacuum) elevation angle e is smaller than the initial elevation angle e_0.

0094-8276/04/2003GL018984

Table 1. Specifications of the ECMWF Datasets Which are Used for the Computation of IMF and VMF

	IMF	VMF
doy 1 in 1979–doy 365 in 2001	2.5° × 2.0° ERA-40 Re-Analysis pressure level dataset (15 levels from 1000 to 10 hPa)	2.5° × 2.0° ERA-40 Re-Analysis pressure level dataset (15 levels from 1000 to 10 hPa)
doy 1 in 2002–doy 238 in 2003	2.5° × 2.0° operational pressure level dataset (15 levels from 1000 to 10 hPa)	2.5° × 2.0° operational pressure level dataset (15 levels from 1000 to 10 hPa)
doy 239 in 2003–now	2.5° × 2.0° operational pressure level dataset (15 levels from 1000 to 10 hPa)	operational pressure level dataset (21 levels from 1000 to 1 hPa, resolution 0.28125°)

six hours, the hydrostatic [*Davis et al.*, 1985] and wet mapping functions as well as the outgoing (= vacuum) elevation angles are determined by raytracing through the pressure levels at ten different initial elevation angles (90°, 70°, 50°, 30°, 20°, 15°, 10°, 7°, 5°, 3.3°). Then, the coefficients a, b and c for the continued fraction form (Equation 1) for the hydrostatic and wet mapping function are estimated in a least-squares procedure. The adjustment shows that three coefficients are enough to map the zenith delays down to 3° elevation with residuals less than 2 mm. So, at each site time series of six parameters (a_h, b_h, c_h, a_w, b_w, c_w) are determined with a resolution of six hours. This approach is only used for test purposes, e.g., for the validation of the fast approach of VMF (see section 2.2).

2.2. Fast Determination of the Coefficient a: VMF(fast)

[8] Although computers are very powerful and fast today, raytracing is still time consuming, especially if it has to be performed for many sites, several (e.g., four) times per day and ten times per grid point. For this reason, a fast version of the rigorous approach has been developed that yields similar values for the mapping functions [*Boehm and Schuh*, 2003] but is about ten times faster than VMF(rig). Instead of determining the raytracing at ten different elevation angles, the raytracing is only calculated for one initial elevation angle of 3.3°. This yields one value for the hydrostatic mapping function, one value for the wet mapping function, and the vacuum elevation angle (~3°). Then, pre-defined formulas are used for the b_h, c_h, b_w and c_w coefficients, and the coefficients a_h, a_w can be determined by simply inverting the continued fraction form (equation (1)).

[9] For the hydrostatic mapping function these coefficients b_h and c_h are taken from the hydrostatic part of the Isobaric Mapping Function IMF. If φ is the geodetic latitude, the coefficients are determined by $b_h = 0.002905$ and $c_h = 0.0634 + 0.0014 \cdot \cos(2\varphi)$ when φ is the latitude. For the wet part of VMF(fast), the coefficients b_w and c_w are constant for all latitudes, because changes in a_w are sufficient to model the dependence of the wet mapping functions on latitude. They are taken from NMF ($\varphi = 45°$: $b_w = 0.00146$ and $c_w = 0.04391$).

[10] The procedure above shows that VMF(fast) is some kind of extension to smfw3 of IMF. The ratio of the wet path delay along a straight line at 3.3° elevation and its zenith delay (smfw3) is replaced by the 'quasi-true' mapping function value at 3.3° initial elevation angle. In contrast to IMFw (smfw3), the bending of the ray is taken into account, and the number of vertical steps is increased considerably for the computation (from 15 to about 1000) by interpolation in height between the pressure levels.

[11] To check the quality of VMF(fast), the values of the mapping functions have been compared to VMF(rig) for CONT02. This is a continuous 15-days VLBI campaign with eight stations in October 2002. (More information about CONT02 is available at http://ivscc.gsfc.nasa.gov/cont02). For an elevation angle of 5° the RMS differences are about 5 mm for the hydrostatic and 1 mm for the wet part when the hydrostatic and wet zenith delays are assumed to be 2000 mm and 200 mm, respectively. This would imply that the corresponding error in the station height is about 2 mm when the cutoff elevation angle is set to 5° [*Niell et al.*, 2001]. Moreover, the RMS differences vanish at about 3° because VMF(fast) is tuned for this elevation angle, and above 5° elevation the RMS differences are also decreasing.

2.3. Mapping Function Parameters Provided by IGG

[12] The Institute of Geodesy and Geophysics (IGG) at the University of Technology, Vienna, gets regular access to data from the ECMWF (European Centre for Medium-Range Weather Forecasts). Derived parameters which are necessary to determine IMF and VMF for non-commercial purposes have been provided to the scientific community since September 2003. Table 1 gives an overview of the ECMWF datasets which are used for these computations. The parameters for IMF (z200 and smfw3) are provided on a global grid with a resolution of 2.5° × 2.0°. The coefficients a_h, a_w for the hydrostatic and wet part of VMF(fast) are determined for all geodetic VLBI stations. This list will be extended to other selected sites, e.g., the IGS (International GPS Service) stations. All parameters are given every six hours and they can be found at the webpage http://www.hg.tuwien.ac.at/~ecmwf.

3. Validation of the VMF

[13] Improved mapping functions are expected to improve geodetic accuracies. Good measures for the quality of geodetic results are the 'baseline length repeatability' and

Table 2. Input Parameters for Hydrostatic and Wet Mapping Functions of NMF, IMF, and VMF(fast)

	NMF	IMF	VMF
hydrostatic	doy, h, φ	z200, φ, h	h, a, (b, c from IMF)
wet	φ	smfw3, h	a, (b, c from NMF)

φ is the station latitude, h is the station height, and doy is the day of the year.

Table 3. Fraction of Baselines With Repeatabilities Better Than With NMF (in %)

	CONT02	IVS-R1	IVS-R4
IMF	54%	79%	80%
VMF(fast)	70%	78%	74%

A clear majority of the baselines is improved with the mapping functions based on NWM.

Table 4. Mean Values of the Improvements in % (δ)

	CONT02	IVS-R1	IVS-R4
IMF	1.9%	9.5%	4.5%
VMF(fast)	5.6%	11.2%	4.4%

The improvement is largest for the IVS-R1 sessions and the VMF(fast) (∼11%).

the difference in baseline length when changing the cutoff elevation angle ('elevation angle cutoff test'). For the geodetic VLBI analysis, the classical least-squares method (Gauss-Markov model) of the OCCAM 5.1 VLBI software package [*Titov et al.*, 2001] is used. Free network solutions with a minimum of squared station coordinate residuals [*Koch*, 1999] are calculated for the 24 h sessions with five Earth orientation parameters being estimated (nutation, dUT1 and pole coordinates). Atmospheric loading parameters are obtained from Petrov, L., and J. P. Boy, (Study of the atmospheric pressure loading signal in VLBI observations, submitted to *J. Geophys. Res.*, 2003.), ocean loading parameters from *Scherneck and Bos* [2002] using the CSR4.0 model by *Eanes* [1994], and total gradient offsets are estimated every 6 hours using the model by *Davis et al.* [1993]. An overview of the input parameters for NMF, IMF and VMF is given in Table 2.

3.1. Baseline Length Repeatabilities

[14] Baseline length repeatabilities are determined for CONT02 (2002, October 16–31), and for all IVS-R1 and IVS-R4 sessions through August, 2003. IVS-R1 and IVS-R4 are weekly 24 h sessions observed on Mondays and Thursdays, respectively, starting in January 2002 [*Nothnagel and Steinforth*, 2003]. For the following investigations all baselines which include the station Tigo Concepcion (Chile) are excluded, because due to the small antenna dish (6 m diameter), the low SNR does not allow a reliable validation of the tropospheric mapping functions. Also, the baselines with station Gilmore Creek (Alaska, U.S.A.) are not considered before the Earthquake on November 3, 2002. The cutoff elevation angle was set to 5°.

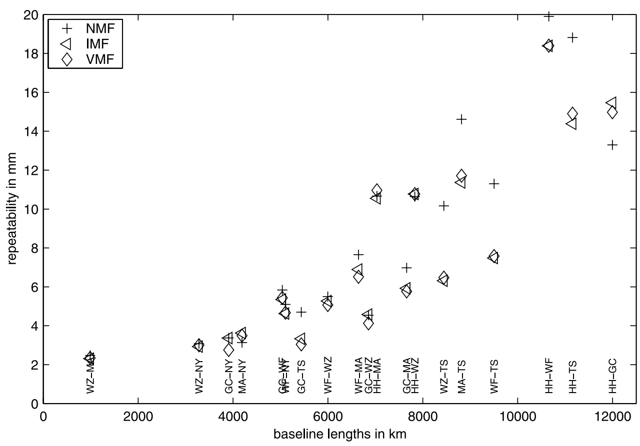

Figure 2. Baseline lengths repeatabilities for IVS-R1. For almost all baselines a clear improvement can be seen with the new mapping functions IMF and VMF. For the specification of the baselines the IVS 2-letter codes are used (see http://ivscc.gsfc.nasa.gov/org/ns.html). The longest baseline from Hartebeesthoek (South Africa) to Gilmore Creek (Alaska, U.S.A.) shows a clear degradation with IMF and VMF, possible reasons are mentioned in the text.

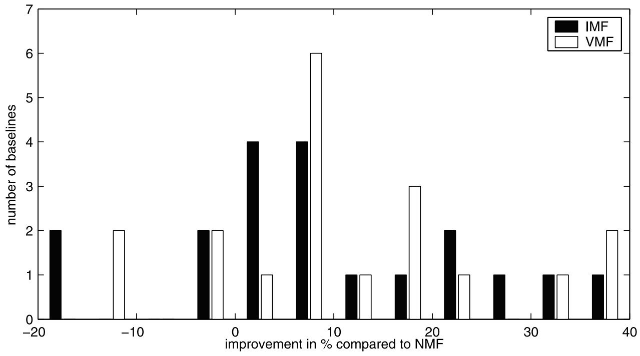

Figure 3. Histogram for the improvements in % of the IVS-R1 baselines with IMF and VMF(fast) compared to NMF. The baseline length repeatability is worse for four baselines only, whereas a clear majority of baselines is improved with the new mapping functions.

[15] Tables 3 and 4 provide information about the improvement of the baseline length repeatabilities of the CONT02, IVS-R1 and IVS-R4 sessions with VMF(fast) and IMF compared to NMF. Table 3 gives the percentage of improved baselines, and table 4 provides the mean value of the improvement over all baselines (equation (2)).

$$\delta = \frac{\sum_{i=1}^{N} \frac{\left(\sigma_{NMF,i} - \sigma_{VMV,i}\right)}{\sigma_{NMF,i}}}{N} \cdot 100 \qquad (2)$$

δ mean improvement over all baselines (i = 1, .., N)
σ repeatability (RMS of the baseline lengths w.r.t. a linear trend)

[16] Figure 2 shows that nearly 80% of the IVS-R1 baseline lengths repeatabilities are improved with IMF and VMF(fast) compared to NMF. Only the longest baseline of about 12000 km from Hartebeesthoek (South Africa) to Gilmore Creek (Alaska, U.S.A.) has a clear degradation in repeatability. Either, this is due to bad numerical weather models at the two sites, or this is simply due to the bad geometry, because only few observations contribute to the determination of the baseline. Thereto, further investiga-

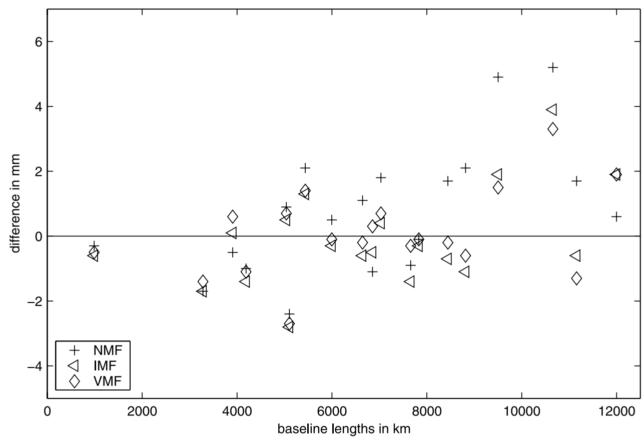

Figure 4. Elevation angle cutoff test for IVS-R1 (10° minus 5°). It can be seen that nearly all differences from NMF are above those from IMF and VMF(fast).

Table 5. RMS Differences Over All Baselines When Changing the Cutoff Elevation Angle From 10° to 5° (in mm; smaller is better)

	CONT02	IVS-R1	IVS-R4
NMF	3.0	2.1	3.7
IMF	2.6	1.5	3.7
VMF(fast)	3.1	1.3	4.2

tions need to be done. Figure 3 shows a histogram for the improvements of the IVS-R1 baselines. It reveals that the clear majority of the baselines (15) is improved and the repeatabilities of four baselines only are degraded with IMF and VMF(fast) with respect to NMF.

3.2. Elevation Angle Cutoff Tests

[17] Another measure for the quality of mapping functions can be obtained by elevation angle cutoff tests [*Herring*, 1983]. These tests show how baseline lengths change when the cutoff elevation angle is varied. They are a good measure for the absolute accuracy of the mapping functions, because if there is a systematic error in the mapping functions the baseline lengths will be influenced when the cutoff angle is changed. In Figure 4, the differences in baseline lengths are shown for the IVS-R1 sessions when changing the cutoff elevation angle from 5° to 10°.

[18] Figure 4 shows that almost all differences for NMF are above those for IMF and VMF(fast). This implies that the baseline lengths with a cutoff elevation angle of 5° are systematically longer with IMF and VMF(fast) than with NMF, since the baseline lengths hardly differ when the cutoff elevation angle is set to 10°. This might be due to systematic effects in either of the mapping functions, and further investigations have to be performed to reveal the actual error sources.

[19] Table 5 summarizes the elevation angle cutoff tests for CONT02, IVS-R1 and IVS-R4. It shows the RMS differences over all baselines when changing the cutoff elevation angle from 10° to 5°. Although modern mapping functions like IMF and VMF have their strengths especially at low elevations, there are no clear improvements or degradations with IMF and VMF, which is a confirmation that the quality of NMF is already rather good.

4. Conclusions and Outlook

[20] Recent mapping functions such as IMF and VMF based on data from numerical weather models like ECMWF provide better repeatabilities of baseline lengths. There is a mean improvement of 5% to 10% versus results obtained by NMF, but still, further investigations remain to be done. Especially, the quality of the mapping functions at certain stations has to be evaluated by means of nearby radiosonde data or different numerical weather models, and a closer look needs to be taken at the systematic effects in the elevation angle cutoff tests.

[21] So far, baseline length repeatabilities are not significantly better with VMF compared to IMF, although VMF exploits the NWM more rigorously. But as the time series get longer, one might speculate that the advantages of VMF become visible, since starting with doy 239 in 2003 the vertical and horizontal resolution of the ECMWF data has increased significantly (to 21 levels and down to 0.5°, respectively), and the quality of ECMWF data will improve steadily in the future.

[22] **Acknowledgments.** We acknowledge the Zentralanstalt fuer Meteorologie und Geodynamik (ZAMG), Vienna, for providing regular access to the ECMWF data. We are also grateful to A. E. Niell for very informative discussions about NWM and mapping functions.

References

Boehm, J., and H. Schuh (2003), Vienna Mapping Functions, in *Proceedings of the 16th Working Meeting on European VLBI for Geodesy and Astrometry*, Leipzig, May 9–10, 2003, pp. 131–143, Verlag des Bundesamtes für Kartographie und Geodaesie.

Davis, J. L., T. A. Herring, I. I. Shapiro, A. E. E. Rogers, and G. Elgered (1985), Geodesy by Radio Interferometry: Effects of Atmospheric Modeling Errors on Estimates of Baseline Length, *Radio Sci.*, *20*(6), 1593–1607.

Davis, J. L., G. Elgered, A. E. Niell, and C. E. Kuehn (1993), Ground-based measurement of gradients in the "wet" radio refractivity of air, *Radio Sci.*, *28*(6), 1003–1018.

Eanes, R. J. (1994), Diurnal and semidiurnal tides from TOPEX/POSEIDON altimetry, *Eos Trans. AGU*, *75*(16), 108.

Herring, T. A. (1983), Precision and accuracy of intercontinental distance determinations using radio interferometry, Ph.D. Thesis, Massachusetts Institute of Technology, Cambridge, MA. (Also published as Air Force Geophysics Laboratory report AFGL-TR-84-0182.)

Herring, T. A. (1992), Modelling atmospheric delays in the analysis of space geodetic data, in *Symposium on Refraction of Transatmospheric Signals in Geodesy*, Netherlands Geod. Commis. Ser. 36, edited by J. C. DeMunk and T. A. Spoelstra, pp. 157–164, Ned. Comm. voor Geod., Delft.

Koch, K. R. (1999), *Parameter Estimation and Hypothesis Testing in Linear Models*, Springer, Berlin.

Marini, J. W. (1972), Correction of satellite tracking data for an arbitrary tropospheric profile, *Radio Sci.*, *7*(2), 223–231.

Niell, A. E. (1996), Global mapping functions for the atmosphere delay at radio wavelengths, *J. Geophys. Res.*, *101*(B2), 3227–3246.

Niell, A. E. (2001), Preliminary evaluation of atmospheric mapping functions based on numerical weather models, *Phys. Chem. Earth*, *26*, 475–480.

Niell, A. E., A. J. Coster, F. S. Solheim, V. B. Mendes, P. C. Toor, R. B. Langley, and C. A. Upham (2001), Comparison of Measurements of Atmospheric Wet Delay by Radiosonde, Water Vapor Radiometer, GPS and VLBI, *J. Atmos. Oceanic Technol.*, *18*, 830–850.

Nothnagel, A., and C. Steinforth (2003), Analysis Coordinator Report, in *International VLBI Service for Geodesy and Astrometry 2002 Annual Report*, edited by N. R. Vandenberg and K. D. Baver, pp. 68–69, NASA/TP-2003-21619.

Scherneck, H.-G., and M. S. Bos (2002), Ocean Tide and Atmospheric Loading, in *International VLBI Service for Geodesy and Astrometry 2002 General Meeting Proceedings*, edited by Nancy R. Vandenberg and Karen D. Baver, pp. 205–214, NASA/CP-2002-210002.

Titov, O., V. Tesmer, and J. Boehm (2001), Occam Version 5.0 Software User Guide, *Auslig Technical Report 7*, AUSLIG, Canberra.

J. Boehm and H. Schuh, University of Technology, Vienna, Austria. (johannes.boehm@tuwien.ac.at)